

PATHWAYS: System Basis and Manuscript Cross-check (Defense Cheat Sheet)

Evidence date: 2026-10-05 (Asia/Manila) · **Code inspected:** `master @ af06a67` (PR #42, merged 2026-10-05) with deltas from `476af5b` traced · **Defense:** 2026-10-07 per the team; the repo's runbook and readiness audit say **2026-10-06** (`docs/runbook-defense-demo.md:7`, `docs/audit-pathways-defense-readiness-20261005.md:4`), so confirm the date.

Status labels. **VERIFIED** means read in code, and a test was run or a recomputation done in this review. **DOCUMENTED ONLY** means stated in docs, with code not run or not found. **PARTIAL** means some of it works or is verified. **UNKNOWN** means not traceable. Manuscript rows use **SUPPORTED**, **PARTIAL**, **ABSENT** or **CANNOT VERIFY**.

Manuscript caveat. The rev-2026 manuscript (DOCX/PDF) is **not in the repository**. Documentary checks use `IDEA.md`, which is "distilled from the rev-2026 capstone manuscript (Chapters 1 to 3)" (`IDEA.md:8`). They also use the use-case disposition in `docs/cr-pathways-doc-reconciliation-2026-10-01.md:84-134`. Exact manuscript wording, table and page numbers are therefore **CANNOT VERIFY** until the file is supplied (see §11).

Citation key. `Mxxxx` means `apps/api/prisma/migrations/<xxxx_*>/migration.sql`, so `M0000` is the baseline through 0026. `API/` means `apps/api/src/modules/`, `WEB/` means `apps/web/src/` and `SH/` means `packages/shared/src/monitoring/`. Roles are SA (System Administrator), PO (Project Officer), ME (M&E Officer), PJM (Project Manager), PgM (Program Manager) and GM (Grant Manager).

Disclose early in the defense: all demo records are **synthetic** seed data. Names come from generated pools and codes follow `BEN-<project>-NNN` (`apps/api/prisma/local-demo-data.ts`). A content test forbids words such as "demo", "synthetic" or "sample" from appearing on screen (`local-demo-data.ts:1698-1712`). The data therefore *looks* real but is not. Say this out loud.

1. One-page quick answer

What it is. PATHWAYS is a web-based, metadata-driven **project information management system with rule-based decision support** for humanitarian and development organizations (`IDEA.md:198`; `README.md`). It sits beside collection tools such as KOBO and exchanges data by **file import**, not live API sync (`IDEA.md:59,309`). **VERIFIED**

Who it serves. Six internal roles: SA, PO, ME, PJM, PgM and GM (`API/auth/rbac-contract.json` roles). PgM and GM see **aggregate-only** beneficiary data (`API/auth/authorization-policy.ts:461`). External stakeholders only read the approved public tracker. Beneficiaries are data subjects and never log in (`IDEA.md:315`). **VERIFIED**

Data-to-decision workflow.

1. Project setup: profile, activities, budget envelope, team.
2. Forms are configured and published by a second person (maker-checker).
3. Data arrives by direct entry or CSV/XLSX/PDF import, followed by mapping, value-map, validation and processing into beneficiaries, enrollments, participations and journey events.
4. Indicators are defined with baseline, target and direction, and measured.
5. The system computes descriptive metrics on dashboards, SADDD and analytics.
6. Deterministic **IF-THEN rules** evaluate 11 catalog metrics, open **alerts** and clone **predefined recommendations**.
7. A **human** reviews them and records an outcome.
8. Reports (CSV/XLS/XLSX/PDF) are generated, and a curated public snapshot is published after a distinct approver signs off.

Technology and design (substantiated).

- Monorepo using pnpm 11.20 and Node 22.
- **Next.js 15 / React 19** web app (`apps/web/package.json`).
- **NestJS 10** modular API with **Prisma 6.19.2** (`apps/api/package.json`).

- **PostgreSQL 17** (Supabase) with row-level security and SECURITY DEFINER SQL functions, plus Supabase Auth (TOTP MFA, aal2 required) and private Storage (`supabase/config.toml:41,306-308`; `apps/api/src/common/guards/supabase-auth.guard.ts:129-133`).
- ECharts and MapLibre for charts and maps. SheetJS, PapaParse and unpdf for imports.
- Vitest and Playwright for tests, and GitHub Actions CI. **VERIFIED** (dependency files)
- Design rationale **documented** only as SDD §1 principles: backend-authoritative authorization, RLS isolation, provenance, metadata-driven forms, trusted metrics, deterministic rules, private by default and PostgreSQL portability (`docs/sdd-pathways.md:14-24`). **No alternatives analysis is documented.** Any "why X over Y" answer is the team's own reasoning; say so (**NOT DOCUMENTED**).

What it computes. All of these are descriptive: counts, shares, **baseline-to-target progress** %, KPI averages, budget utilization, timeline elapsed / remaining / overdue days, activity completion, milestone on-time %, SADDD disaggregation with small-cell suppression, survey pre/post means and splits, and weighted evaluation totals. **No predictive model, forecasting, impact estimate or AI** exists (`IDEA.md:313`; audit `docs/audit-pathways-manuscript-alignment-20261001.md` SL-19). **VERIFIED**

How alerts and recommendations are produced.

- A System Administrator configures project rules over a closed catalog of 11 metrics. The operators are `LT`, `LTE`, `EQ`, `GTE`, `GT`, `BETWEEN`, with nested AND/OR groups (`API/rules/rule-contract.ts:12-24,42`).
- A scheduled drain and sweep evaluates them **in SQL** (`M0060:142-312,317-448`).
- A TRUE result opens one alert per episode and clones the rule's 1-10 **predefined** recommendation texts (`M0059:465-486`).
- FALSE auto-resolves the episode. Missing data (UNAVAILABLE) never opens or closes an alert (`M0059:492,518-523`).
- Humans review the alert and record Accept, Partially accept, Decline or Escalate. **No text is generated.** **VERIFIED** (327 rule tests passed and synthetic cases recomputed, §5-6)

Most important limits.

- **Hosted scheduling** is Partly met (G-F10-7). Alerts may need a manual drain (`docs/prd-pathways.md:1192`; DR-06).
- The system **does not evaluate individual beneficiaries**. Survey results are project-level, released by period and suppressed under 5 pairs (`IDEA.md:310`; §8).
- **No in-app path** writes survey pre/post scores (`assessment_results`) or evaluation scores. Both are seed-only (`docs/deferred-features.md:56`; `apps/api/prisma/local-demo-seed.ts:31-33`).
- **Four different "budget utilization" formulas** are shown in different places (§5, P0-2).
- Monitoring-dashboard **participation and enrollment counts are withheld** (`SENSITIVE_RELEASE_NOT_ENABLED_V1`, `M0000:2045-2079`).
- R4 project templates are not built, the forty-hour saving is unmeasured, and the **ISO/IEC 25010 UAT has not been run**, so Objective 3 has no results (`docs/qad-pathways.md:527`).

2. Source register and currency check

Source	Version / status	Date (last change)	Use here	Currency note
Manuscript rev-2026 (Ch. 1-3, Table 5 matrix, Tables 6-31 use cases)	Not in repo	-	Through IDEA.md and the reconciliation CR	Blocker for verbatim checks (§11 P0-6)
IDEA.md (manuscript distillation)	Locked 2.0	Reconciled 2026-10-01	Objectives 1.1-3.8, P1-P8 / R1-R8, scope (IDEA.md:154-237, 303-317)	Current; says repository wins on conflicts (:8)
docs/prd-pathways.md	Locked 2.0 (tables updated)	2026-10-05	13 features, 100 gates, UC-F*, FR/NFR, matrix §3.3 (:89-130)	Most current status authority
docs/sdd-pathways.md	Locked 2.0	2026-10-04	Architecture, principles	Drift: says no indicator library, no journey notes, no scheduler (:298, 300, 116), but all three exist
docs/qad-pathways.md	Locked 2.0	2026-10-05	Test IDs, UAT instrument	§3.5 still lists gates as Not met that the PRD marks Met (:315, 332)
docs/cr-pathways-doc-reconciliation-2026-10-01.md	Applied	2026-10-03	UC001-UC026 disposition (:84-134)	Historical; its Not-met lists are superseded
docs/audit-pathways-manuscript-alignment-20261001.md	Findings	2026-10-04	MA-01 to MA-18, trace table	Trace table is stale (e.g. 1.2 "no archive route", R7 "no budget metric")
docs/audit-pathways-defense-readiness-20261005.md	Active, highest	2026-10-05	DR-01 to DR-09	Lists this cheat sheet as "Missing"
docs/deferred-features.md	Control	2026-10-05	Gaps register	Current
docs/state.md	Control	2026-10-05 (one line)	Release state	Stale body ("0058-0060 not applied to hosted")
docs/runbook-defense-demo.md, docs/handoff-defense-demo-seed.md	Working	2026-10-05 / 10-04	Demo seed	Says coverage map empty (superseded) and CSV risk (register says resolved)
Code apps/, packages/, 38 migrations 0000-0063	af06a67	2026-10-05	Implementation authority	Local checkout 476af5b is 47 commits behind; 0063 and the PDF renderer exist only on master
RBAC API/auth/rbac-contract.json + authorization-policy.ts	amendments to 0055	-	Role grants	JSON and TS match (scripted comparison)
Tests: 373 tracked test files + 50 SQL runtime suites	-	-	Evidence	See §11 "Checks run"

Use-case IDs from the prompt (cr-pathways-doc-reconciliation-2026-10-01.md:105-110):

Manuscript UC	Name	PRD use case	Disposition
UC018 (Table 23)	Review Rule-Based Alerts	UC-F10-1, UC-F10-2	Supported; named-individual notification is not carried forward
UC019 (Table 24)	Review Rule-Based Project Recommendations	UC-F11-1, UC-F11-2	Was Partly supported; Auto-resolved has since been built (0058/0059)
UC020 (Table 25)	Generate Reports	UC-F12-1	Supported
UC023 (Table 28)	Configure Evaluation and Alert Parameters	UC-F10-3	Supported; overlap warning not established

Conflicting revisions to resolve before the defense:

1. The QAD §3.5 Not-met list vs the PRD Met statuses.
2. The alignment-audit trace table vs its own closed findings.

3. Recommendation statuses: - The reconciliation CR says New / Reviewed / Resolved / Dismissed. - The PRD state machine says New / Reviewed / AutoResolved. - The schema enum has all five values, but f10 rows only use NEW / REVIEWED / AUTO_RESOLVED (M0059:81-94).
4. The SDD's "missing is never zero" (:471) vs the approved zero-display CR.
5. The defense date: 10-06 vs 10-07.

3. Feature-by-feature basis

3a. Basis, screen, role, status

Feature (PRD)	User need (fishbone / P → R)	Objective · matrix · use case	Screen → API · roles	Status
Auth, RBAC, workspace (F1)	P1 dependence on technical staff → R1	1.8 · F1 · UC001-005	/login, /auth/mfa, /settings/users → SupabaseAuthGuard plus @RequirePermission · all roles; users managed by SA/PjM/PgM	VERIFIED (code; CSV-RBAC tests run) · 312-grant runtime suite DOCUMENTED ONLY (docs/qad-pathways.md:385-389)
Project profile (F2)	P5 scattered records → R5	1.2 · F2 · UC006	/projects, /projects/new → POST/PATCH /projects, archive · create/update PjM only; archive SA, PjM	VERIFIED (code); program create granted to no role (seed only)
Activities, milestones, proof, overdue explanation, extensions (F2)	P7 delays not visible → R7	1.2, 1.5 · F2 · UC007	/projects/:id/activities/:activityId → activities, extension-requests · create PjM; progress PO/ME/PjM (assigned); proof review ME; extension verify ME, decide PjM	VERIFIED; milestones read-only (milestones.manage granted to no one)
Budget and expenses (F2)	Budget evidence (1.2)	1.2 · F2 · UC007	/projects/:id/budget → finance · budgets PjM/PgM/GM; expense submit PO/ME/PjM → verify ME → approve PjM → sign-off PgM/GM, distinct actors	VERIFIED (M0034:242-280,754-796)
Beneficiary profile (F3)	P5, P6 → R5	1.3 · F3 · UC010, UC012	/beneficiaries (+new/edit/duplicates) → beneficiaries · PO/ME/PjM records; detail needs step-up (fresh TOTP ≤ 15 min); PgM/GM aggregate only	VERIFIED; no merge (link/keep-distinct audit only); archive granted to no one; PIN UI hidden
Journey tracking (F4)	P6 participation hard to track → R6	1.3 · F4 · UC013, UC014	/projects/:id/journey-stages, beneficiary detail → participants · stages SA/ME/PjM; events and corrections PO/ME/PjM	VERIFIED; corrections are append-only
Forms (F5)	P2 repeated handling, P3 manual setup → R2, R3	1.4, 2.1 · F5 · UC009	/collection/forms → metadata · build SA/ME; author cannot publish; direct entry PO/ME	VERIFIED; published forms change by new version only
Imports and mapping (F6)	P2 → R2	1.4, 2.1 · F6 · UC011	/collection/import → imports · upload/validate/process SA/PO/ME; manual map SA/ME	VERIFIED; generic rows lack beneficiary linkage (§7)
Indicators and library (F7)	P3, P4 → R3, R4	1.5 · F7 · UC008	/projects/:id/indicators, /indicators/library · SA/ME/PjM	PARTIAL: only one live DERIVED recipe; R4 templates not built
Monitoring dashboard + SADDD (F8)	P3 → R3	1.6, 2.2, 2.3 · F8 · UC015, UC017	/dashboard, /analytics → dashboards · SADDD SA/ME/PjM/PgM/GM	PARTIAL: SADDD closed-period only; participation counts withheld
Descriptive analytics, survey (F9)	P7 → R7	1.6, 2.2 · F9 · UC016	/analytics views (timeline, survey, participation, budget, trends)	PARTIAL: survey data seed-only; trend has ≤ 1 point per indicator
Rule-based alerts (F10)	P7 → R7	1.7, 2.4 · F10 · UC018, UC023	/alerts, /alerts/repository (rules board) → rules · rules SA only; alerts all six roles	PARTIAL: hosted schedule Partly met (G-F10-7)
Recommendations (F11)	P7 → R7	1.7, 2.4 · F11 · UC019	/recommendations/:id (list redirects to /alerts) · all six roles	VERIFIED (code and tests)
Reports and visualization (F12)	P3 → R3	1.6 · (no manuscript F#) · UC020, UC021, UC022	/reports/* → reports · 6 kinds × CSV/XLS/XLSX/PDF; beneficiary summary PO/ME/PjM	VERIFIED (local); designed PDF hosted check pending
Coverage map (F12)	Visibility	1.6	/analytics "map" → GET /analytics/project-map (projects.read)	DOCUMENTED ONLY + code read; CR says local QA only

Feature (PRD)	User need (fishbone / P – R)	Objective · matrix · use case	Screen → API · roles	Status
Public tracker (F13)	P8 limited external visibility → R8	1.8 · manuscript F12 · UC025, UC026	/transparency (staff), /public/projects (anonymous) · submit/approve/publish SA/PJM/PgM/GM; approver ≠ submitter	PARTIAL : G-F13-5 hosted not verified
Project evaluation (OECD-DAC weights)	Evaluation outputs	1.6	GET /evaluation ; criteria init SA; weights SA/ME	PARTIAL : no create/score/submit/sign-off endpoint; demo EHK evaluation is seeded
Backups, labels	Admin	SL-13 · UC024	/settings/backups placeholder; labels redirect	ABSENT (UC024 not carried forward)

3b. Inputs, transformation, storage, outputs, limits

Feature	Inputs	Transformation and validation	Stored (main tables)	Outputs / next module	Limits and evidence
Projects	title, dates, status, PHP envelope (≤ 2 dp), area, partners (≤ 20)	DTO validation; optimistic concurrency (expectedUpdatedAt , 409); envelope replace = archive old row and insert new	projects , project_budget_records (PROJECT_PROFILE_TOTAL), user_project_assignments	Activities, overview tiles, rules (timeline)	Archive is soft; archived projects leave all scopes. API/projects/projects.service.ts:217-259, 519-606
Activities	plan dates, status, assignments; proof 1-10 files + SHA-256; session count 0-100000	ME reviews proof; APPROVE at 100% means COMPLETED; extension approval overwrites planned_end_date (no history)	project_activities , activity_updates , evidence_media , activity_extension_requests	Timeline and overdue metrics, rule recompute trigger	API/activities/activities.service.ts:1583-1597, 2188-2361; API/activity-extensions/activity-extensions.service.ts:423-464
Expenses	amount > 0, receipt ≤ 10 MiB (signature checked)	PENDING → VERIFIED → APPROVED, plus a separate sign-off; distinct actor per stage	budget_expense_entries , expense_signoffs	Budget metrics (APPROVED only)	API/finance/finance.service.ts:24-62; M0034:242-280, 754-796
Beneficiaries	registration form values, both consents, guardian for minors, age ≥ 5	form contract check, then domain rules; duplicate code or identifier → 409 REVIEW	beneficiaries , enrollments, identifiers, consent records, form_submissions	SADDD, journey, follow-up rule	API/beneficiaries/beneficiaries.service.ts:193-330, 727-1093
Journey	stage (ENTRY/CORE/BRANCH/FOLLOW_UP), event, correction + reason	chronological order enforced; correction = new row with correctsEventId	journey_stages , beneficiary_journey_events	Journey progress (browser), recipe M08d (gated)	API/participants/participants.service.ts:150, 581, 692-707
Forms	fields (1-100, 8 data types, lowercase codes)	publish checks registration contract and activity binding; versioning +1	forms , form_versions , form_fields	Direct entry, imports	API/metadata/metadata.service.ts:458-520, 1124-1157
Imports	CSV/XLSX/XLS/text-PDF ≤ 5 MiB, ≤ 5000 rows	worker-thread parse; AUTO_SMART_V2 mapping; value map ≤ 50 entries; type coercion; duplicate-key flag; VALID/INVALID per row; chunked process	data_import_batches/rows , metadata_mappings → registration, participation or generic submission	Beneficiaries, participations, journey	packages/imports/src/limits.ts:1-21; API/imports/imports.service.ts:834-1056, 1877-1985
Indicators	mode (MANUAL/DERIVED), numericKind, direction, baseline, target, period, type label	measurement must match the definition period; corrections chain; semantic fields immutable	project_indicators , project_indicator_measurements	KPI, progress, rules (MANUAL or derived activity % only)	SH/metric-contract.ts:82-153; API/indicators/indicators.service.ts:439-508
Rules	metric, operator, threshold(s), AND/OR tree, severity, 1-10 recommendations	config validated (domains, depth ≤ 4, ≤ 32 conditions); edit = new DRAFT version; activate archives the prior version	alert_rules (conditions_json), alert_rule_recommendations , internal cursors and evaluations	Alerts, recommendations, notifications on outcome	API/rules/rule-contract.ts:42-194; M0031:3840-3918
Reports	kind, project, format	permission-gated source read; suppression carried through; designed PDF via headless Chromium, else pdfkit	reports + private artifact	Download / export	API/reports/reports.service.ts:41-48, 449-467, 540-551

Feature	Inputs	Transformation and validation	Stored (main tables)	Outputs / next module	Limits and evidence
Public tracker	curated summary fields only	CHECK limits the snapshot to id, title, code, approvedSummary, area, sector, dates; approver ≠ submitter	public_project_snapshots, publication_requests	Anonymous pages	M0034:86-90,144-147; API/public/public.service.ts:151-198

4. System architecture and workflow

Stack (from dependency files; **VERIFIED**):

Layer	Technologies
Web	Next ^15.2.2, React ^19, TanStack Query and Table, ECharts ^5.6, MapLibre 6.10, react-hook-form + zod, Radix/shadcn, Tailwind ^3.4
API	NestJS ^10.4.15, Prisma 6.19.2, zod, class-validator, helmet, pino, pdfkit 0.20.2, puppeteer-core 25.12 + @sparticuz/chromium (designed PDF, master only), Sentry
Data	PostgreSQL 17 (Supabase) with RLS, SECURITY DEFINER functions, Supabase Auth (TOTP) and private Storage bucket pathways-private
Imports	SheetJS xlsx 0.20.3, PapaParse 5.7, unpdf 1.8 (packages/imports/package.json:26-31)
Hosting	Vercel (DOCUMENTED ONLY: docs/ops-pathways.md:14-23); Dockerfiles exist. CI runs prisma validate, migration replay, secret grep, lint, typecheck, test and build (.github/workflows/ci.yml). Playwright e2e is not in CI.

Patterns actually used:

- A modular NestJS monolith of 24+ modules (apps/api/src/app.module.ts).
- Global auth guard plus per-route @RequirePermission. A route without a permission gets 403 (common/guards/supabase-auth.guard.ts:140-144).
- Defense in depth: every request is a transaction that sets app.organization_id and app.user_id (apps/api/src/prisma/prisma.service.ts:125-215), and RLS policies call p05_has_project_permission (M0000:1171-1203).
- Business rules in SQL functions: p34_* for expenses, p06_* for metrics, f10_* for rules.
- Shared zod contracts in packages/shared.
- @Roles() is unused, so RolesGuard is a no-op (inference from the code; harmless).

Documented rationale vs inference. Documented: the SDD §1 principles (docs/sdd-pathways.md:14-24) and the "deterministic rules, no AI" scope (IDEA.md:313). The following are **inferences, not documented**: SQL-side metrics are there "so numbers cannot be altered by the browser", and Next/Nest were chosen for TypeScript end-to-end. Label them as the team's reasoning.

Data flow:

```
Forms / Imports —validate+map—> beneficiaries, enrollments, participations, journey events, form_submissions
Projects / Activities / Budgets / Expenses / Indicators+measurements —> SQL metric functions (p06_*, p10_f9_*, p34_*)
  ↳ Dashboards (home, role overview, monitoring, SADD), Analytics (timeline, survey, participation, budget, trends, map)
  ↳ Reports (6 kinds × 4 formats) —> Public snapshot (curated fields only, approved by a 2nd person)
  ↳ Rules: source change marks project PENDING → drain (5 min) / sweep (hourly) → evaluate in SQL
        → alert episode + cloned recommendations → human review/outcome → notifications (on outcome only)
```

Permissions model.

- Role and permission come from the database grants plus the TS ceiling (`authorization-policy.ts:453-459`).
- Project scope (`API/auth/authorized-data.service.ts:12-33`):
- SA covers the whole organization.
- PgM covers assigned projects plus programs they manage.
- Every other role covers explicit assignments only.
- Organization configuration: indicator library, rule templates, users, rules calendar (Asia/Manila, SQL only).
- Project configuration: profile, activities, budget, journey stages, forms, indicators, project rules (copied from a template), evaluation weights and publication.

Error and empty states (examples):

- Metrics use a `MetricCell` with state AVAILABLE, ZERO, MISSING, NOT_APPLICABLE or SUPPRESSED plus a reason. Missing is never silently 0 in the API (`SH/metric-contract.ts:30-47`). The web **does** show 0 for some missing cells (approved in `docs/cr-pathways-overview-zero-display.md`).
- Analytics returns 503 on a provider fault, never an empty payload (`API/analytics-insights/analytics-insights.service.ts:112-121`).
- Empty lists say so, e.g. "No activities match the current view" and "No indicators configured".

5. Calculation reference

General rules:

- Server decimals are exact ($\text{bigint} \times 10^4$) and rounded **once**, half away from zero ([SH/metric-math.ts:17-55](#)).
- The business date is Asia/Manila.
- Small cells of 1-4 are suppressed.
- "Progress" is **baseline-to-target movement, not a success rating**, and is never clamped ([SH/metric-math.ts:65](#)).
- Classes: **Out** = output or activity completion, **Proc** = process or efficiency, **Oc** = outcome-level, project aggregate only. **No impact metric exists**.

5a. Calculation table

ID	Label · where	Inputs (provenance)	Formula	Period · rounding	Edge cases	Cls · status	Evidence
M01	KPI achievement % · project overview tile	each non-archived indicator's progress cell (M09)	$KPI = (1/r) \cdot \sum \text{progress}_i$ over the r indicators with a visible progress	as-of now, any period · 1 dp	0 indicators → MISSING NO_INDICATORS; $r = 0 \rightarrow$ NO_MEASUREMENT; missing never counted as 0; not clamped	Proc/Oc · VERIFIED	SH/overview-metrics.ts:103-122
M02	Budget utilization % · overview tile	planned = PROJECT_PROFILE_TOTAL envelope only; spent = $\sum$ APPROVED expenses	$U = \text{spent} \div \text{envelope} \times 100$	all-time · 1 dp	no envelope → NO_PLANNED_BUDGET; spent 0 → NO_APPROVED_EXPENSES (UI shows "0%"); envelope 0 → NOT_APPLICABLE; PENDING, VERIFIED and REJECTED excluded	Proc · VERIFIED	SH/overview-metrics.ts:125-138 ; API/projects/project-overview-metrics.service.ts:47-73
M03	Efficiency ratio · Budget tab	M01, M02	$E = KPI\% \div \text{Budget}\%$	2 dp	either input missing → INPUT_UNAVAILABLE; budget 0 → NOT_APPLICABLE	Proc · VERIFIED	SH/overview-metrics.ts:141-157
M04	Beneficiaries reached / target · overview	SADDD total (M21), projects.target_beneficiaries	reached = distinct enrolled individuals overlapping the project period	released only after the project closes	open project → MISSING; 1-4 → SUPPRESSED; UI shows "0" when missing	Out · VERIFIED	API/projects/project-overview-metrics.service.ts:81-94
M05	Timeline % · overview, directory	start, end, business date	$(\text{today} - \text{start} + 1) \div (\text{end} - \text{start} + 1) \times 100$, clamped 0-100	1 dp	missing or invalid dates → MISSING	Proc · VERIFIED	SH/overview-metrics.ts:159-177
M07	KPI% / Budget% per project · PjM/PgM/GM dashboards	M01, M02 inputs	$\text{Budget}\% = \text{Math.round}(100 \cdot \text{spent} \div \text{envelope})$	integer (browser)	$\text{envelope} \leq 0 \rightarrow \text{"—"}$; shown capped "100% (x% over budget)"	Proc · VERIFIED (code)	WEB/features/dashboard/role-overview/project-metrics-reads.tsx:16-31
M08	Indicator current value (MANUAL) · "Actual"	latest uncorrected measurement for the exact definition period	latest by recorded_at, then id (no sum, no average)	definition period	none → MISSING NO_MEASUREMENT (list shows 0)	Oc/Out · VERIFIED	M0000:1398-1405 ; API/indicators/indicators.service.ts:87-147
M08g	Derived: Activity completion % (the only live recipe)	activities due in the indicator period, not cancelled or archived	$100 \cdot \text{COMPLETED} \div \text{eligible}$	4 dp	0 eligible → NOT_APPLICABLE ZERO_DENOMINATOR	Out · VERIFIED	M0000:1491-1495

ID	Label · where	Inputs (provenance)	Formula	Period · rounding	Edge cases	CIs · status	Evidence
M08a-f	Derived recipes: participation count, distinct attending, attendance per person, journey events, form sum/avg	participations, journey events, VALIDATED submissions	count / distinct / ratio / sum / avg	4 dp	withheld on dashboards (<code>SENSITIVE_RELEASE_NOT_ENABLED_V1</code>) but computed in the Indicator Summary report	Out · PARTIAL	M0000:1409-1490,1887-1913; M0034:399-517; docs/deferred-features.md:62
M09	Indicator progress % · indicator list, KPI chart, rule metric	actual, baseline, target, direction	$P = (\text{actual} - \text{baseline}) \div (\text{target} - \text{baseline}) \times 100$	4 dp, not clamped	DESCRIPTIVE or missing baseline/target → NOT_APPLICABLE; target = baseline → ZERO_DENOMINATOR; wrong-direction target → DIRECTION_CONFLICT	Proc/Oc · VERIFIED	SH/metric-math.ts:66-93
M10	Indicator trend points · Analytics	latest measurement per indicator period	value series	≤ 50 indicators × 120 points	measurement must equal the definition period, so ≤ 1 point per indicator (degenerate trend)	Out · PARTIAL	API/analytics-insights/insights-math.ts:48-65
M11	Indicator summaries (count, mean, min, max by unit and kind)	visible current values	mean/min/max per (unit, numerickind)	4 dp	different units never averaged together	Out · VERIFIED (code)	API/dashboards/descriptive-analytics.ts:62-87
M14-15	Activity and milestone state counts and shares	current status of items due in the period	count per state; share = cell ÷ Σ	4 dp	undated items excluded; uses current status, not status at period end	Out · VERIFIED (code)	M0028:122-132; API/dashboards/descriptive-analytics.ts:45-60
M16-19	Participation records, attending individuals, enrolled records/individuals · monitoring, "Beneficiary reach"	participations, enrollments	counts	-	always overwritten to MISSING (<code>SENSITIVE_RELEASE_NOT_ENABLED_V1</code>); SA "Project monitoring" cards show 0	Out · PARTIAL	M0028:95-139; override M0000:2045-2079; WEB/features/dashboard/role-dashboard.tsx:314-326,479
M21-25	SADDD total; sex, age-band, disability, completeness marginals	INDIVIDUAL beneficiaries enrolled in the project period, current profile	distinct counts; age = completed years at project end: 0-9, 10-14, 15-17, 18-24, 25+, Unknown	closed project period only; one release, then frozen	any cell of 1-4 suppresses the whole table; a later source change → STALE (MISSING RESTATEMENT_REVIEW_REQUIRED)	Out · VERIFIED (code; SQL suites documented)	M0028:141-432; M0000:1244-1256
M27-31	Timeline elapsed %, remaining days, overdue days, activity completion %, overdue activities · Analytics timeline (same functions as the rule metrics)	project dates, activity status and planned end	$\text{elapsed} = \max(0, \text{today} - \text{start}) \div (\text{end} - \text{start}) \times 100$; $\text{remaining} = \text{end} - \text{today}$; $\text{overdue} = \max(0, \text{today} - \text{end})$; $\text{completion} = \text{COMPLETED} \div \text{eligible} \times 100$; $\text{overdue count} = \text{non-completed with planned_end} < \text{today}$	4 dp; elapsed not clamped	project not PLANNED/ONGOING → NOT_APPLICABLE; any non-completed activity with no date → overdue count MISSING	Proc · VERIFIED (tests run)	API/rules/rule-metrics.ts:144-234,384-406; M0045:185-248
M32	Milestones on time %	COMPLETED milestones with both a target and a completion date	$\text{onTime} \div \text{rated} \times 100$ (same day counts as on time)	1 dp	rated 0 → MISSING	Proc · VERIFIED (tests)	M0045:231-244; SH/descriptive-analytics.ts:517-537
M33-37	Survey: pairs, mean pre/post %, mean change (pp), improved/same/declined, excluded records	<code>assessment_results</code> PRE/POST (score ÷ max × 100); latest of each per enrollment	$\text{meanPre} = \Sigma \text{pre} \div n$; $\text{meanPost} = \Sigma \text{post} \div n$; $\text{change} = \text{meanPost} - \text{meanPre}$	exactly one defined non-overlapping period · 1 dp	< 5 pairs → suppressed; split complementary-suppressed; PgM/GM need a closed period (frozen copy)	Oc (project-level) · PARTIAL : no in-app writer for <code>assessment_results</code>	M0057:55-238; SH/descriptive-analytics.ts:357-464

ID	Label · where	Inputs (provenance)	Formula	Period · rounding	Edge cases	CIs · status	Evidence
M38	Survey form results (respondents, answered, missing, mean, true/false)	latest VALIDATED TRAINING_SURVEY submission per beneficiary	counts; mean = $\Sigma \div$ answered	4 dp	any count 1-4 → whole report suppressed	Out/Oc · VERIFIED (code)	M0034:900-987
M39	Participation breakdown (by activity, month, attendance)	participation records , not people	counts	optional period	1-4 hidden plus one complementary cell; PgM/GM get 403	Out · VERIFIED (tests)	API/analytics-insights/insights-math.ts:13-33
M40	Budget summary per currency · Analytics card	planned = Σ all non-archived budget rows (envelope + activity lines); approved; pending = PENDING + VERIFIED	$U = \text{round}(\text{approved} \div \text{planned} \times 1000) \div 10$	1 dp	planned 0 → null; double-counts when activity lines exist	Proc · VERIFIED (known defect)	insights-math.ts:70-95; docs/deferred-features.md:65
M42	"Active budget alerts" · SA dashboard KPI row	NEW + REVIEWED alerts	count of all open alerts	-	label says "budget" but counts every alert	Proc · VERIFIED (code defect)	WEB/features/dashboard/action-kpi-row.tsx:45; API/dashboards/action-counts.service.ts:100-121
M43	Overdue activities + most overdue	planned_end < today, not COMPLETED/CANCELLED	count; daysLate of the oldest	business date	-	Proc · VERIFIED (tests)	action-counts.service.ts:60-85
M51	Project health label (browser)	max open alert severity	CRITICAL → Critical; HIGH/MEDIUM → At risk; else On track; PLANNED → Planned	-	ignores KPI and budget entirely	Proc · VERIFIED (tests)	WEB/features/dashboard/role-overview/health.ts:10-19
M54	Activity beneficiaries reached	Σ beneficiaries_reached_this_session on APPROVED proof updates	sum	-	officer-typed number; detail shows 0 without permission	Out · VERIFIED (code)	API/activities/activities.service.ts:845-873
M56	Activity budget utilization (list)	activity's budget rows	$\text{Math.round}(\text{approved} \div \text{planned} \times 100)$	integer	planned 0 → null	Proc · VERIFIED (code)	activities.service.ts:879-923
M60	Budget module utilization (browser)	rows by activity; project-level row = envelope – Σ activity allocations; APPROVED = used	$U = \text{round}(\text{used} \div \text{allocated} \times 100)$; remaining = allocated – used	integer	allocated 0 → null; ≥ 90 danger, ≥ 70 warning, remaining < 15% warning (visual only)	Proc · VERIFIED (tests)	WEB/features/projects/budget-module/budget-math.ts:37-102, 107-143
M63	Evaluation weighted and overall score	criterion score, max and weight (weights total 100)	$w_i = \text{round}(\text{score}_i \div \text{max}_i \times \text{weight}_i, 4)$; overall = Σw_i	on SUBMITTED	DB triggers only; no API writes scores	Oc (project) · PARTIAL	M0000:3722-3741, 3953-3977; API/evaluations/evaluations.controller.ts:21-56
M65	Journey progression % (browser)	journey PARTICIPATION events	$\text{round}(\text{distinct stages reached} \div \text{configured stages} \times 100)$	integer	directory passes [], so it always shows 0% ; ABSENT still counts as reached	Out · PARTIAL (TODO(BACKEND))	WEB/features/beneficiaries/beneficiary-utils.ts:83-113; WEB/lib/services/pathways-client.ts:2540
M67	Public tracker counts	published snapshots	count, distinct sectors, latest date	-	no indicator, budget or beneficiary numbers are public	Out · VERIFIED (code)	WEB/features/public/public-tracker-view.tsx:19-24
M68-69	Analytics KPI card; activity completion card (browser)	visible progress; activity list	$\text{round}(\text{mean progress})$; completed ÷ total	integer	completion card counts cancelled activities in the total; KPI tone ≥ 70 green	Proc · VERIFIED (code)	WEB/features/analytics/analytics-dashboard.tsx:687-689, 1018-1019, 1081

Four budget utilization formulas (same synthetic project, verified):

Where	Denominator	Example result
Overview tile (M02)	envelope only	30.1%
Analytics card (M40)	envelope + activity lines	21.5%
Budget page (M60)	envelope	30%
Rules metric	all non-archived rows (as M40)	21.5% → can miss a 90% rule

Three timeline formulas: the overview is inclusive and clamped, Analytics and rules are exclusive and unclamped, and the directory uses the overview function. Pick one sentence for the panel (§10 Q9).

5b. Worked synthetic examples (**VERIFIED** : recomputed 2026-10-05 by running the repo's own functions, **npx tsx** from the scratchpad)

Example A: KPI, budget and efficiency (synthetic project "Mock-A").

Indicator	Inputs	Progress
A	baseline 10, target 30, latest 20, HIGHER	$(20 - 10) \div (30 - 10) \times 100 = 50$
B	DERIVED activity completion 75 (3 of 4), baseline 0, target 100	75
C	no measurement	MISSING , not counted
D	LOWER, baseline 12, target 4, value 9	$(9 - 12) \div (4 - 12) \times 100 = 37.5$

- KPI = $(50 + 75 + 37.5) \div 3 = 54.1667$, giving **54.2%** (4 indicators, 3 reported). *Interpretation:* on average, reported indicators have covered 54% of their baseline-to-target distance. This is not a pass/fail score.
- Budget: envelope ₱500,000.00 with approved spend ₱150,500.50. A VERIFIED ₱20,000 expense is excluded. Utilization = **30.1%**.
- Efficiency = $54.2 \div 30.1 = 1.80$, shown as **"1.8"**.
- Analytics card with a ₱200,000 activity line: $150,500.5 \div 700,000 = 21.5\%$ (double count).

Example B: timeline for a project running 2026-01-01 to 2026-12-31, on 2026-10-05.

- Overview: $278 \div 365 = 76.2\%$.
- Analytics/rules: $277 \div 364 = 76.0989\%$.
- Remaining days 87; overdue 0.

Example C: SADD release (synthetic closed project with 30 eligible individuals).

- Age 10-14: 6, 15-17: 7, 18-24: 9, 25+: 8.
- Sex M 16, F 14. Disability with 5, without 25.
- All cells are 0 or ≥ 5 , so the table is released: **total 30**, male share 53.3%.
- If one edit before release makes disability "with" = 4, **every cell is suppressed** (complementary suppression).
- An edit after release makes it **STALE** permanently (**M0028:183-194, 269-342**).

Example D: survey (synthetic, one defined closed period, 7 enrollments).

- Normalized pre/post scores: (50, 75), (60, 80), (40, 40), (70, 60), (50, 90), (30, 60), (50, 70).
- Pairs **7**, mean pre **50.0**, mean post **67.9**, mean change **+17.9 pp**.
- The improved / same / declined split is 5 / 1 / 1. Same = 1 falls in 1-4, so **all three counts are suppressed**.
- The rule metric SURVEY_MEAN_IMPROVEMENT_POINTS gives $125 \div 7 = 17.8571$ (4 dp; a separate implementation).

- *Interpretation:* the cohort average moved; it says nothing about any one person (\$8).

6. Rule reference

Engine facts:

- **Operators** are LT, LTE, EQ, GTE, GT and BETWEEN. LT and GT are strict, LTE and GTE inclusive, BETWEEN inclusive at both ends, and EQ is exact at 4 dp ([API/rules/rule-engine.ts:44-69](#); [M0060:205-208](#)).
- **AND/OR groups:**
 - Depth ≤ 4 , ≤ 32 conditions.
 - Three-valued logic. AND is FALSE if any child is FALSE, else UNAVAILABLE if any child is UNAVAILABLE, else TRUE. OR is the mirror image.
 - Evidence is kept for every condition ([rule-engine.ts:154-165](#)).
- **Severity** is LOW, MEDIUM, HIGH or CRITICAL.
- **Category is not stored.** The UI derives Performance / Schedule / Financial / Delivery / Participation / Outcome from the metric ([WEB/features/rules-board/rule-board-model.ts:16-22,51-72](#)).
- **Scope:**
 - An organization **template** (project_id NULL) cannot be activated. It is copied into a project.
 - At most 20 ACTIVE rules per project ([M0031:3840-3852,3866](#)).
- **Project override = separate rule rows**, so two projects can hold different thresholds with no engine change. *Real seeded example:* SSG **ACTIVITY_OVERDUE_COUNT** **GTE 2** while CRL holds **> 3** (§6a). *Hypothetical:* CRL **BUDGET** **GTE 90** (HIGH) while another project holds **GTE 75** (MEDIUM). Each is evaluated only against its own project's metric.
- **Favorable conditions are possible but framed as alerts.** Any operator works on any metric (verified: **PROJECT_REMAINING_DAYS** **GTE 60** $\rightarrow$ TRUE). However:
 - There is no polarity field and no "info/positive" severity.
 - The alert message is fixed: "Rule conditions matched." ([M0059:467](#)).
 - Every seeded rule is negative.
- **Gap (P1-4).**
- **Metric catalog (11):**

Metric	Formula
INDICATOR_CURRENT_VALUE	indicator's current value
INDICATOR_PROGRESS_PERCENT	M09
PROJECT_TIMELINE_ELAPSED_PERCENT, PROJECT_REMAINING_DAYS, PROJECT_OVERDUE_DAYS	M27-29
ACTIVITY_COMPLETION_PERCENT, ACTIVITY_OVERDUE_COUNT	M30-31
ACTIVITY_OVERDUE_DAYS	days past planned end for one bound activity
BUDGET_UTILIZATION_PERCENT	Σ APPROVED on non-archived rows $\div$ Σ planned of all non-archived rows $\times 100$ (M0060:390-404)
BENEFICIARY_FOLLOW_UP_PERCENT	share of ACTIVE enrollments whose latest participation is NEEDS_FOLLOW_UP, suppressed < 5 (M0060:405-416)
SURVEY_MEAN_IMPROVEMENT_POINTS	avg(post – pre), suppressed < 5 pairs (M0060:417-429)

Indicator metrics need an **eligibility** approval row that **no code writes** (P1-5), so in practice the seeded rules use project, activity and aggregate metrics only.

6a. Seeded rules (synthetic demo; `apps/api/prisma/local-demo-data.ts:1448-1589`)

Rule (project)	IF	THEN alert (severity)	Linked predefined recommendation(s)	Override
OPERATIONS_BOTTLENECK (WSH)	TIMELINE_ELAPSED_% > 50 AND ACTIVITY_COMPLETION_% < 40	HIGH	Hold a recovery planning meeting; Request a no-cost timeline extension	project rule
PROJECT_PAST_END_DATE (WSH)	PROJECT_OVERDUE_DAYS > 0	CRITICAL	Review the closure plan with the donor	project rule
ACTIVITY_DELAYS (SSG)	ACTIVITY_OVERDUE_COUNT ≥ 2	MEDIUM	Reschedule delayed activities; Reassign field support	SSG uses ≥ 2
ACTIVITY_DELAYS_WATCH (CRL)	ACTIVITY_OVERDUE_COUNT > 3	LOW	Discuss workload at the next coordination meeting	same metric, different threshold per project
ACTIVITY_OVERDUE_ANY (CRL)	ACTIVITY_OVERDUE_COUNT ≥ 1	MEDIUM	Confirm a new date for the overdue activity	auto-resolve demo
CLOSING_SOON (ALS)	PROJECT_REMAINING_DAYS < 30	LOW	Start the close-out checklist	project rule
BUDGET_NEAR_EXHAUSTED (CRL)	BUDGET_UTILIZATION_% ≥ 90	HIGH	Review the remaining budget with the Grant Manager; Pause new grant tranches until reviewed	project rule
FOLLOW_UP_GAP (CRL)	BENEFICIARY_FOLLOW_UP_% ≥ 25	MEDIUM	Schedule household follow-up visits	project rule
LOW_SURVEY_IMPROVEMENT (WSH)	SURVEY_MEAN_IMPROVEMENT_POINTS < 20	MEDIUM	Review the hygiene club training content	project rule

Recommendation basis. Recommendations are **predefined per rule version** (`alert_rule_recommendations`, 1-10 each, title ≤ 160 and text ≤ 2000 characters). When an alert episode opens, every one of them is cloned (`M0059:476-486`). There is no generation, no eligibility filtering and no user editing of text after activation. Their status is NEW, then REVIEWED, then AUTO_RESOLVED (system only). An outcome is recorded against the recommendation, but the status does not change: Accept and Partially accept move the **linked alert** to ACTIONED (`M0059:712-716`).

6b. Lifecycle and decision flow (**VERIFIED** from SQL and tests)

```
Source edit (project/activity/indicator/measurement) -> project job PENDING (M0031:4701-4756)
Hourly sweep (all projects with ACTIVE rules; needed for budget/beneficiary/survey and date-driven metrics)
Drain (every 5 min, <=20 projects) -> snapshot -> evaluate ACTIVE rules in SQL (M0060:142-312)
  TRUE  & no open episode -> NEW alert + cloned recommendations (M0059:465-486)
  TRUE  & open episode   -> same alert, revision+1 (deduplicated) (M0059:487-496)
  FALSE                               -> episode closed; open alert -> AUTO_RESOLVED; undecided recs -> AUTO_RESOLVED (M0059:492,501-515)
  UNAVAILABLE (missing) -> nothing changes (M0059:518-523)
Human: Review (NEW->REVIEWED, note) | Outcome ACCEPT/PARTIAL -> ACTIONED; DECLINE/ESCALATE keep state | Resolve/Dismiss
Threshold edit -> new DRAFT version -> activate archives old; old open alerts become SUPERSEDED (not auto-resolved; P1-6)
```

Triggers and who.

- There is **no human "evaluate now" button**. Evaluation runs through machine routes `POST /internal/rules/drain|sweep` with a 64-hex bearer token, and is off unless `RULES_WORKER_ENABLED=true` (`API/rules/rules-machine.controller.ts:9-31`).
- The GitHub Actions cron (`* /5 drain, 7 * * * * sweep`) is gated by `RULES_DISPATCH_ENABLED` (`.github/workflows/rules-dispatch.yml:4-21`).
- Rules permissions (`rules.*`) are **SA only**. `alerts.*` and `recommendations.*` go to all six roles (`rbac-contract.json`).
- Aggregate rules have audience gates. Budget alerts are visible only to holders of `budgets.read` + `expenses.read` (PjM, PgM, GM). Beneficiary and survey alerts need `beneficiaries.records.read` or `assessments.detail.read` (PO, ME, PjM). So the SA who writes a budget rule **cannot see its alerts** (`M0060:119-133`).
- Notifications** are created only when an outcome is confirmed, and are in-app only. No notification is sent when an alert fires (`M0031:3164-3179`; `M0059:748-753`).

Dry run (POST /rules/dry-run, SA). It uses **typed, synthetic inputs only**. It reads no stored project data and writes nothing: no alert, notification or decision ([API/rules/rules-human.service.ts:198-206](#); [WEB/features/analytics/rule-test-workspace.tsx:294-296](#)). It runs the **TypeScript** evaluator, while live runs **PL/pgSQL**. The semantics mirror each other, with two known differences: a missing observation is UNAVAILABLE in TS but raises an error in SQL, and the budget $\geq 1e14$ guard exists only in SQL. No parity test exists (P1-7).

6c. Firing and non-firing cases (VERIFIED: run 2026-10-05 against rule-engine.ts + rule-metrics.ts, synthetic)

Rule	Input	Metric cell	Result	Live effect
BUDGET ≥ 90	approved 95,000 / planned 100,000	95	TRUE	new alert + 2 recommendations
BUDGET ≥ 90	85,000 / 100,000	85	FALSE	open alert → AUTO_RESOLVED
BUDGET ≥ 90	90,000 / 100,000 (boundary)	90	TRUE (GTE inclusive; GT would be FALSE)	fires
BUDGET ≥ 90	no budget records	MISSING EMPTY_POPULATION	UNAVAILABLE	no change, no false alarm
OVERDUE_DAYS > 0	end 2026-09-30, today 2026-10-05	5	TRUE	CRITICAL alert
OVERDUE_DAYS > 0	today = end date	ZERO	FALSE (strict >)	none
OVERDUE_DAYS > 0	project COMPLETED	NOT_APPLICABLE	UNAVAILABLE	none
REMAINING_DAYS ≥ 60 (favorable)	today 2026-07-01	91	TRUE	still raised as an "alert"

What the alert screen shows ([WEB/features/analytics/human-review-workspace.tsx:456-559](#)):

- Rule version, freshness, evaluated time, reporting date and calendar version.
- The condition tree.
- An evidence table of metric, **observed value + unit**, operator + **threshold**, and result.
- Linked recommendation titles and texts.

It does **not** show the indicator's measurement period, and it has **no drill-down to supporting records**: the alert JSON carries no source record IDs (P1-8).

7. End-to-end source-to-output traces

#	Trace	Status
T1	CSV "Sex" column → upload and worker parse (API/imports/imports.service.ts:566-641) → automatic map (:390-441) → validate and type coercion (:834-1018) → process <code>promoteRegistration</code> → <code>beneficiaries.sex</code> , <code>birth_date</code> , enrollment (API/beneficiaries/beneficiaries.service.ts:905-973) → <code>p06_compute_saddd</code> (M0028:141-196) → GET /dashboards/saddd → Analytics SADDL chart and BENEFICIARY_SUMMARY report	VERIFIED (code path); shows only after the project closes
T2	ACTIVITY_MONITORING import → <code>promoteParticipation</code> writes <code>progress_status</code> and a journey event (API/participants/participants.service.ts:430-463) → rule metric BENEFICIARY_FOLLOW_UP_PERCENT (M0060:405-416) → hourly sweep → alert FOLLOW_UP_GAP → "Schedule household follow-up visits"	VERIFIED (code); depends on the hosted scheduler
T3	Expense submit → verify (ME) → approve (PJM) (M0034:754-796) → APPROVED amount → M02 overview, M60 budget page, M40 analytics, rule BUDGET_NEAR_EXHAUSTED	PARTIAL: the denominators differ (§5)
T4	Indicator measurement (MANUAL) → M08 → M09 progress → M01 KPI → efficiency M03; source op enqueues a rule re-evaluation (M0031:901-917)	VERIFIED; indicator rules blocked by missing eligibility writer
T5	Activity proof approved at 100% → COMPLETED → M30 completion, M31 overdue count → OPERATIONS_BOTTLENECK / ACTIVITY_OVERDUE_ANY → auto-resolve when cleared (demo stage apps/api/prisma/defense-demo-stage-autoresolve.ts:11-66)	VERIFIED (code, seed); hosted not verified
T6	Report generate → designed PDF (Chromium, API/report-pdf/report-pdf.renderer.ts) or pdfkit fallback → export	VERIFIED locally (CR docs/cr-pathways-report-pdf-renderer.md:5); hosted pending

Broken or unverified links.

- **B1.** Generic imported survey rows have no beneficiary or enrollment, so they never reach the survey report or the FORM_* recipes (`imports.service.ts:1920-1957` ; `M0034:320-321,922-927`).
- **B2.** Nothing in the app writes `assessment_results` (survey pre/post, rule SURVEY_*) or evaluation scores.
- **B3.** Monitoring participation and enrollment counts are withheld.
- **B4.** Derived recipes are masked in the UI but computed in the report.
- **B5.** The hosted scheduler is Partly met.
- **B6.** The public pages are not verified on hosted (G-F13-5).

8. Beneficiary and evaluation limits

What the data can show (project level only):

- Who is enrolled and registered, with demographics.
- Attendance and participation records per activity.
- Journey events: stages reached, follow-ups and corrections.
- Officer-reported session counts.
- Cohort survey means and an improved / same / declined split, for ≥ 5 pairs in a defined period.
- SADDD distributions after project close.

What it cannot show:

- It cannot show that an individual **learned, improved, succeeded or failed**. The manuscript limitation says it does not evaluate beneficiaries as individuals (`IDEA.md:310` , SL-18 Met).
- Attendance or stage progression **is an output, not an outcome**.
- The survey view is a cohort mean of normalized scores, with no individual display. Aggregate-only roles see only frozen closed-period releases, and differencing is blocked (`docs/prd-pathways.md:1087-1103` ; QAD-A15 to A21).
- Nothing is causal: there is no comparison group, no attribution and no impact.

Evidence needed for a stronger claim:

1. Recorded, validated pre/post instruments per enrollment, which needs an **in-app writer for** `assessment_results` (absent).
2. Defined learning or skill indicators with baselines and targets (OUTCOME type) measured over time.
3. Follow-up outcome records (employment, school retention and the like) linked to enrollments.
4. A comparison or baseline cohort for any attribution claim.
5. Data-quality checks: instrument validity, and minimum n per subgroup.

Until then, phrase claims as **"project-level outcome monitoring"** (audit MA-15).

Step-up for beneficiary detail. It requires a fresh TOTP (≤ 15 min) or a session-bound PIN grant (bcrypt, 5-try lockout), enforced on the server and audited (`API/auth/beneficiary-step-up.ts:3-54` ; `M0037:47-198`). It replaced a hard-coded browser PIN (`docs/cr-pathways-beneficiary-step-up.md`). The PIN **UI is hidden** (`docs/deferred-features.md:10`).

9. Manuscript-to-implementation cross-check

Status values: SUPPORTED, PARTIAL, ABSENT or CANNOT VERIFY. Objective text is from [IDEA.md:202-237](#).

Item	Claim (short)	Status	Evidence	Smallest accurate revision (if needed)
1.1	Web metadata-driven system centralizing project, activity, beneficiary and progress data	SUPPORTED	§3, §4	-
1.2	Project profile, activity tracking, budget evidence	SUPPORTED	archive route exists; milestones are read-only	"...milestones (viewed; created by setup)"
1.3	Beneficiary profile and journey to "encode, organize, monitor and evaluate beneficiary participation..."	PARTIAL	no merge; no individual evaluation	Replace "evaluate beneficiary participation" with " monitor beneficiary participation"
1.4	Configure forms, map, validate, process	SUPPORTED	§3b, T1	-
1.5	Configure activities, indicators, targets	SUPPORTED	-	-
1.6	Dashboards, descriptive analytics, SADD, progress summaries, visualization; " identify trends and outcomes "	PARTIAL	trends ≤ 1 point; participation counts withheld	"...summarize current performance and project-level outcome indicators"
1.7	Alerts for underperforming indicators, delayed timelines, beneficiary progress issues, low assessment improvement; " suggest actions "	PARTIAL	metrics exist; indicator rules need an eligibility row; survey data is seed-only; suggestions are predefined	"... present predefined recommendations configured by administrators"
1.8	Secure RBAC and public tracker	PARTIAL	hosted public check not done (G-F13-5)	-
2.1	Metadata-driven preparation	SUPPORTED	packages/imports , metadata module	-
2.2	Descriptive analytics using target-vs-actual, completion, participation, assessment, budget, timeline	PARTIAL	four budget formulas; participation counts withheld	add: "budget utilization = approved ÷ planned envelope"
2.3	SADD by sex, age and disability	SUPPORTED	M21-25	add: "released after project close; cells < 5 suppressed"
2.4	If-then rules for delays, underperformance, budget, beneficiary progress, missing follow-up , low assessment improvement	PARTIAL	follow-up metric is the share flagged NEEDS_FOLLOW_UP, not "missing follow-up records"	"...beneficiaries flagged as needing follow-up"
3.1-3.8	ISO/IEC 25010 UAT	ABSENT (results)	"UAT has not been run" (docs/qad-pathways.md:527)	report as planned instrument; present unit/integration evidence instead
PD (prescriptive analytics)	"prescriptive analytics proposing courses of action"	ABSENT	MA-02	" rule-based decision support with predefined recommendations "
PD (beneficiary outcome assessment)	improves outcome assessment	PARTIAL	project-level survey only	" project-level outcome monitoring " (MA-15)
PD (40 hours)	removes ~40 h of preparation	CANNOT VERIFY	no timing evidence (MA-03)	"estimated by staff; to be measured in UAT"
R1, R2, R5	structured access, organized preparation, centralized records	SUPPORTED	§3	-
R3	outputs with less manual configuration	PARTIAL	see 1.6	-
R4	reusable project structures	PARTIAL	indicator library only; templates deferred	"reusable indicator definitions (library); project templates future work"

Item	Claim (short)	Status	Evidence	Smallest accurate revision (if needed)
R6	participation and progression tracking	SUPPORTED	journey, notes (0049)	-
R7	summaries, rule flags, action guidance	PARTIAL	hosted scheduler	-
R8	controlled public visibility	PARTIAL	snapshot only; no indicators or budget public	"approved summary fields (no indicator values yet)"
Fishbone (People, Process, Data, Technology)	P1-P8	SUPPORTED as context	IDEA.md:143-167	-
UC018	Review alerts	SUPPORTED	UC-F10-1/2	note "in-app notification on outcome only"
UC019	Review recommendations	SUPPORTED	auto-resolve built	outcome is recorded; status stays New/Reviewed
UC020	Generate reports	SUPPORTED	6 kinds × 4 formats	-
UC023	Configure alert parameters	SUPPORTED	SA only; no overlap warning	"System Administrator configures..."
Literature: "activity completion alone does not show outcomes"	design intent	SUPPORTED	outputs and outcomes are kept separate	-
Exact table/page wording	-	CANNOT VERIFY	manuscript file missing	-

Contradictory wording to fix (application vs manuscript):

- PRD UC-F10-2 names PjM and PgM as outcome actors, but **all six roles hold alerts.outcome.record**. Pick one: change the grants, or restate the use case.
- The rules-board preview says the alert goes "for assigned project users" (**rule-board-model.ts:152**), but no notification fires on alert creation.

UI vs decisions it should support:

Decision the manuscript expects	What the current UI provides
"Which projects are at risk?"	Health label from alert severity, but KPI and budget are ignored (M51)
"Is spending on track?"	Budget tile; formulas inconsistent
"Which activities are late?"	Overdue count + most overdue + required explanation. Good
"Who needs follow-up?"	Count-level alert only. Correct by privacy design : staff open the beneficiary list

10. Likely defense questions and short answers

1. **What is the design approach?** A metadata-driven, modular monolith. Forms and fields are data, not code. Authorization is server-side with RLS, and all metrics come from SQL functions under deterministic rules ([docs/sdd-pathways.md:14-24](#)). There is no AI.
2. **Why this stack?** It is TypeScript end to end: Next.js for the UI and NestJS for the API, with PostgreSQL row-level security for tenant isolation and Supabase for auth (TOTP) and storage. The rationale is documented only as SDD principles; the comparative reasons are ours.
3. **Walk a role workflow.** A PO uploads a CSV and the ME maps and validates it. Then a PO records proof, the ME verifies it, the PjM approves expenses and the PgM/GM signs off. Distinct actors are enforced in SQL ([M0034:242-280,754-796](#)).
4. **Where does a dashboard number come from?** From the API's SQL function over committed records. The browser recomputes only a few display values (budget page, role cards), which are listed in §5.
5. **How does mapping work?** AUTO_SMART_V2 suggests columns. The ME can map manually with type and value maps (≤ 50). Validation coerces types and marks each row VALID or INVALID, and invalid rows stay staged ([API/imports/imports.service.ts:683-1018](#)).
6. **How is KPI achievement computed?** It is the mean of the indicators' $(\text{actual} - \text{baseline}) \div (\text{target} - \text{baseline}) \times 100$ over the indicators that have data. Missing indicators are excluded, not counted as zero. Example A gives 54.2%.
7. **Is 100% "success"?** No. It means the target was reached. Values over 100 are shown capped, with an overrun note. The contract states "no universal KPI score or success threshold" ([API/dashboards/dashboard-contract.ts:63](#)).
8. **Why do two budget figures differ?** The Analytics card and the rule metric divide by the envelope plus the activity lines, while the overview and budget page divide by the envelope. This is a registered defect ([docs/deferred-features.md:65](#)). Quote the overview figure.
9. **Why do two timeline % differ?** The overview counts days inclusively and clamps the value. Analytics and rules count exclusively and can exceed 100 after the end date.
10. **How are rules configured per project?** The SA copies a template into the project and sets its own threshold. Each project's rows are evaluated only on that project's metrics; CRL and SSG hold different overdue thresholds.
11. **Can rules flag good performance?** Yes, syntactically: e.g. GTE on progress. They are still raised as "alerts", with no positive severity. Improving this is listed future work.
12. **Are recommendations AI-generated?** No. Each rule carries 1-10 fixed texts, and these are cloned when an alert opens. A human records Accept, Partially accept, Decline or Escalate with a note ([M0059:476-486](#)).
13. **What if data is missing?** The metric becomes UNAVAILABLE with a reason. An alert is never opened or closed on missing data, and missing is never treated as 0 in the API.
14. **What if data is updated or backdated?** Source edits queue a re-evaluation. If the condition clears, the alert auto-resolves. A threshold edit creates a new version, and old alerts show "superseded". A SADDD release frozen earlier becomes STALE.
15. **Can you say a beneficiary learned?** No, only that the cohort mean changed, for ≥ 5 pairs in a defined period (§8).
16. **How is privacy protected?** Through org and project scope on every request, plus RLS. Beneficiary detail needs step-up, PgM/GM are aggregate-only, cells under 5 are suppressed, and the public snapshot is restricted by a database CHECK.
17. **How did you verify it?** By the evidence in §11 "Checks run": 1,000+ unit tests passed in this review, plus SQL runtime suites documented in the QAD. Be candid that **UAT has not been run**.
18. **Does it replace KOBO?** No. It imports files from such tools, and has no API sync by design ([IDEA.md:59](#)).
19. **Who can change thresholds?** Only the System Administrator ([rules.*](#)). This is audited as [rule.draft](#) / [rule.activate](#) ([M0031:3916-3918](#)).
20. **What is the evaluation score 87.55?** It is a seeded OECD-DAC weighted sum, computed by database triggers: $\Sigma \text{score} \div \text{max} \times \text{weight}$. **There is no in-app scoring screen** (DR-08).

11. Gaps and corrections

P0: factual or demo blockers (fix the wording or the demo before the defense)

ID	Type	Issue	Smallest safe fix / wording	Owner / location
P0-1	Demo	Hosted alerts depend on a scheduler that is Partly met (DR-06)	Run one manual <code>rules-dispatch</code> drain before the session, or say "evaluation runs on a schedule; shown here after a drain"	<code>docs/runbook-defense-demo.md:51-53</code>
P0-2	App	"Budget utilization" differs by screen (30.1% / 21.5% / 30%)	Demo the overview and budget page only; say the analytics/rule denominator fix is registered	<code>insights-math.ts:70-95</code> ; <code>M0060:390-404</code>
P0-3	App	SA KPI card "Active budget alerts" counts all alerts	Rename the label to "Open alerts", or avoid that card	<code>WEB/features/dashboard/action-kpi-row.tsx:45</code>
P0-4	Manuscript	"Prescriptive analytics", "evaluate beneficiaries" and "outcome assessment" claims	Use the §9 revisions	Manuscript Ch. 1 Purpose (MA-02, MA-15)
P0-5	Demo	devV2 reseed and <code>--verify</code> not recorded; no frozen tag (DR-04)	Reseed, record <code>--verify</code> , tag the build	<code>scripts/db/defense-demo.mjs</code>
P0-6	Source	Manuscript file not in repo	Supply the rev-2026 DOCX/PDF: Chapters 1-3, Table 5 matrix, Tables 23-25 and 28	Team
P0-7	Disclosure	Demo data looks real, and evaluation and survey scores are seeded	State "synthetic seed data" at the start	-

P1: incomplete behavior (honest "future work" answers)

ID	Issue	Fix	Location
P1-1	No in-app writer for survey pre/post (<code>assessment_results</code>) or evaluation scores	add write paths under <code>evaluations.*</code> and an assessment form promotion	<code>API/evaluations/</code> ; <code>docs/deferred-features.md:56</code>
P1-2	Monitoring participation and enrollment counts withheld; Beneficiary reach shows "Not available" / 0	enable the release after privacy review	<code>M0000:2045-2079</code>
P1-3	Derived recipes masked except activity completion; the report computes them (inconsistent)	align the report gate with the dashboard	<code>M0000:1887-1913</code> ; <code>M0034:399-517</code>
P1-4	No favorable/positive rule polarity or "info" severity	add a polarity field and message template	<code>API/rules/rule-contract.ts:179</code> ; <code>M0059:467</code>
P1-5	Indicator rule metrics need an eligibility row that nothing writes	add an approval UI or route	<code>M0031:3736-3744</code>
P1-6	Superseded alerts are never auto-resolved and their actions are hidden	close or resolve on version change	<code>M0031:3295, 4429-4431</code>
P1-7	Dry-run (TS) vs live (SQL) evaluators lack a parity test	add a parity suite	<code>API/rules/rule-engine.ts</code> ; <code>M0060</code>
P1-8	Alert evidence has no drill-down to source records and no measurement period	add record links	<code>M0031:3383-3405</code>
P1-9	Journey progress always 0% in the directory	pass participations, or compute on the server	<code>WEB/lib/services/pathways-client.ts:2540</code>
P1-10	Milestones, programs, beneficiary/form/indicator archive: permissions granted to nobody	grant or descope	<code>authorization-policy.ts:50-108</code>
P1-11	Hosted scheduler (G-F10-7), public pages (G-F13-5), designed PDF on hosted	human activation steps	<code>docs/deferred-features.md:51, 66</code> ; <code>PRD:1460</code>
P1-12	R4 project templates; 40-hour measurement; UAT	CR or descope; run UAT	<code>docs/deferred-features.md:38, 39, 44</code>

P2: polish

ID	Issue	Location
P2-1	Analytics completion card counts cancelled activities	analytics-dashboard.tsx:1081
P2-2	"This month" counts start at 00:00 UTC, not Manila	API/dashboards/role-overview.service.ts:173-250
P2-3	Survey aggregate does not exclude dummy/archived beneficiaries	M0057:94-147
P2-4	Stale docs: SDD drift, QAD §3.5, audit trace table, state.md, runbook CSV/map notes, CR headers	docs/
P2-5	Health label ignores KPI and budget; the UI tone thresholds (70, 60, 80) are visual only	health.ts; analytics-dashboard.tsx:1019

Honest to demonstrate on 10-06/07 (local seed verified, 23/23 checks per docs/handoff-defense-demo-seed.md:15):

- Projects and activities, with proof review and overdue explanation.
- The expense verify → approve → sign-off chain.
- Beneficiary registration and import mapping.
- Journey events.
- Indicators and the KPI tile.
- SADDD on the closed EHK project.
- The survey view (seeded) and alerts and recommendations, after a drain.
- Reports in all four formats, and the public tracker.

Later work: P1 items, UAT, hosted scheduler and public check.

Checks run in this review (2026-10-05, cloud container, no database, no hosted target touched)

Command (workspace)	Result
pnpm --filter @pathways/shared test	10 files, 108/108 passed
pnpm --filter @pathways/config test	2 files, 19/19 passed
pnpm sad:test	47/47 passed
apps/api: npx vitest run src/modules/rules	14 files, 327/327 passed
apps/api: vitest on rules, reports.service, dashboards unit tests, analytics-insights, auth CSV-RBAC	21 files, 410/410 passed (includes the 327 rule tests); analytics.service.test.ts and report-artifact.test.ts failed to load : papaparse was not installed because the network blocked the SheetJS CDN
apps/api: vitest on project-overview-metrics, indicators, finance ledger, activity-progress, role-overview, action-counts	248/248 passed (two targeted runs, 127 + 121)
apps/web: vitest on features analytics, dashboard, rules-board, reports and lib/rbac	42 files, 472/472 passed
apps/web: vitest on budget-math, beneficiary-utils, project-detail-view, reporting periods, health, action-kpi-row, pinned-charts	7 files, 41/41 passed
Independent recomputation, npx tsx verify.ts with repo functions on synthetic inputs	KPI 54.2, budget 30.1, efficiency 1.8, timeline 76.2; 8 rule cases as in §6c

Command (workspace)	Result
Not run	<code>packages/imports</code> tests (dependency blocked), <code>*.local.test.*</code> , 50 SQL runtime suites, Playwright e2e, any hosted step

Test runs were on the `476af5b` checkout. The metric and rule source is unchanged on `af06a67` except `packages/shared/src/monitoring/project-map.ts`.

12. Two-minute demo script (synthetic seed; not executed in this review)

Prerequisites:

- The PjM account is assigned to SSG and WSH (check the seed's assignments).
- devV2 or local reseeded and `--verify` clean.
- One `rules-dispatch` drain done.
- Sign in with TOTP.
- Say: "All records are synthetic seed data."

Time	Role · screen	Show	Say (one line)
0:00	PjM · <code>/dashboard</code>	Role overview: projects, KPI%, budget% (capped), overdue	"Each number comes from SQL over approved records; missing data is never zero."
0:20	PjM · SSG <code>/projects/:id/activities</code>	Overdue activity, explanation required	"Overdue = planned end before today and not completed."
0:35	PjM · SSG budget tab	Envelope, used (APPROVED only), remaining, verify → approve → sign-off	"Utilization = approved ÷ envelope."
0:55	PjM · <code>/alerts</code>	An open alert, e.g. WSH PROJECT_PAST_END_DATE (no seeded decision) → evidence table (metric, observed, threshold, result) → predefined recommendation	"If project overdue days > 0 THEN critical alert; the text is predefined, not generated."
1:15	PjM · record outcome	Accept with a note → alert ACTIONED	"A human decides; the outcome is audited and notified in-app."
1:30	ME · <code>/analytics</code> EHK (closed)	SADDD by sex, age band and disability	"Released after close; any cell of 1-4 suppresses the table."
1:45	ME · <code>/reports</code>	Generate a Project Summary PDF	"Six report kinds in four formats."
1:55	Public · <code>/public/projects</code>	Approved snapshot only	"No beneficiary, budget or indicator data is public."

If no populated project exists, the minimum synthetic records needed are:

- One organization and one user per role.
- One ONGOING project with dates and a ₧ envelope.
- 3 activities, one overdue.
- One budget line and 2 APPROVED expenses.
- One MANUAL indicator with baseline, target and one measurement.
- One CLOSED project with ≥ 5 enrolled individuals (≥ 5 in every SADDD cell).
- One active rule (`ACTIVITY_OVERDUE_COUNT GTE 1`) with one recommendation.
- One published snapshot.